CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 18-2023: A 19-Year-Old Woman with Dyspnea and Tachypnea

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PRESENTATION OF CASE

Dr. Jacquelyn M. Nestor (Medicine): A 19-year-old woman was transferred to this hospital because of respiratory failure.

The patient had been in her usual state of health until 3 weeks before the current presentation, when pain developed in the right shoulder and the right flank. She was evaluated in the emergency department of another hospital and was discharged home after testing for kidney stones and appendicitis was reportedly negative.

Two weeks before the current presentation, dyspnea with exertion and nonproductive cough developed. Six days after the onset of dyspnea and cough, the patient departed on a vacation cruise with her family. During the 8-day trip, dyspnea and cough worsened and prevented her from participating in activities such as swimming and hiking. By the end of the cruise, she had severe dyspnea with walking, and she required the use of a wheelchair during her trip home. After returning to Massachusetts, she presented to the emergency department of the other hospital for evaluation.

The patient reported ongoing cough and progressively worsening dyspnea but no chest pain or hemoptysis. The temporal temperature was 37.1°C, the heart rate 132 beats per minute, the blood pressure 133/47 mm Hg, the respiratory rate more than 40 breaths per minute, and the oxygen saturation 88% while she was breathing ambient air. She appeared ill and had increased work of breathing. A murmur was heard on cardiac examination; other details of the examination are not available. An electrocardiogram (ECG) reportedly showed sinus tachycardia and lateral and precordial T-wave inversions. Urinalysis results and blood levels of glucose, electrolytes, and troponin T were normal, as were the results of liver-function and kidney-function tests; the blood level of N-terminal pro–B-type natriuretic peptide (NT-proBNP) was 24,913 pg per milliliter (reference range, 5 to 125). Other laboratory test results are shown in Table 1. Blood cultures were obtained.

Computed tomography (CT) of the chest, performed after the administration of intravenous contrast material according to a pulmonary embolism protocol, report-

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Table 1. Laboratory Data.

Variable	Reference Range, Adults, Other Hospital	On Presentation, Other Hospital	Reference Range, Adults, This Hospital*	On Presentation, This Hospital
Hemoglobin (g/dl)	12.0–16.0	9.1	12.0–16.0	8.4
Hematocrit (%)	36.0–48.0	31.5	36.0–46.0	28.4
Mean corpuscular volume (fl)	79.0–98.0	66.7	80.0–100.0	66.8
White-cell count (per μl)	3500–11,000	15,540	4500–11,000	15,690
Differential count (%)				
Neutrophils	35–66	72	—	—
Lymphocytes	25–45	20	—	—
Monocytes	0–13	6	—	—
Eosinophils	0–8	1	—	—
Platelet count (per μl)	150,000–400,000	745,000	150,000–400,000	620,000
Albumin (g/dl)	3.5–5.2	2.9	3.3–5.0	3.1
Lactate (mmol/liter)†	0.5–2.0	0.2	0.5–2.2	1.3
Erythrocyte sedimentation rate (mm/hr)	—	—	0–20	77
C-reactive protein (mg/liter)	—	—	<8.0	147.8
Arterial blood gases				
Fraction of inspired oxygen	—	—	—	0.70
pH	—	—	7.35–7.45	7.51
Partial pressure of carbon dioxide (mm Hg)	—	—	35–42	31
Partial pressure of oxygen (mm Hg)	—	—	80–100	206

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.
† To convert the values for lactate to milligrams per deciliter, divide by 0.1110.

edly revealed pericardial and pleural effusions and multifocal lung opacities but no evidence of pulmonary emboli. Point-of-care ultrasonography reportedly revealed a left ventricular ejection fraction of 25%, marked aortic regurgitation, and a possible aortic-root abscess. Supplemental oxygen was administered through a high-flow nasal cannula. Treatment with intravenous furosemide, vancomycin, ceftriaxone, and azithromycin was started. The patient was transferred to the cardiac intensive care unit of this hospital.

On arrival at the cardiac intensive care unit, the patient received bilevel positive airway pressure. A family member accompanying the patient provided additional history. The patient had had intermittent back pain beginning 6 months before the current presentation. Two months before the current presentation, she had had asymptomatic infection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Five weeks before the current presentation, she had had a dental cleaning. During the cruise, the patient had two episodes of diarrhea without fever or chills; the family member did not know whether other persons on the cruise had been ill. The patient had told her family that she had noticed visible pulsations in her chest during the 2 weeks before the current presentation, and the family member reported that the patient's breathing had appeared labored while she was sleeping.

Other medical history included anxiety and anemia. The patient took supplemental iron and had no known drug allergies. She did not use alcohol, tobacco, or other substances. She lived with her family in Massachusetts and worked in child care. There was no family history of cardiac or autoimmune disease.

The temporal temperature was 36.8°C, the heart rate 118 beats per minute, the blood pressure 145/58 mm Hg in the right arm and 136/65 mm Hg in the left arm, the respiratory rate 48 breaths per minute, and the oxygen saturation 100% while the patient was receiving bilevel positive airway pressure (fraction of inspired oxygen, 0.70). The body-mass index (the weight in kilograms divided by the square of the height in meters) was 21.6. While the patient was sitting in an upright position, she had increased work of breathing and intermittent cough. The heart rhythm was tachycardic, with a grade 2/6 systolic murmur at the base and a grade 3/4 decrescendo diastolic murmur at the base that radiated to the apex, without an S_3 gallop. Carotid and subclavian bruits were present. The jugular venous pressure was 8 cm of water. There were rales and decreased breath sounds in both lungs. Brachial, ulnar, radial, dorsalis pedis, and posterior tibial pulses were symmetric. The remainder of the examination was normal.

The blood levels of glucose, electrolytes, thyrotropin, lipase, human chorionic gonadotropin, and troponin T were normal, as were the results of tests of liver function and kidney function. A urine toxicology screening test was negative. The erythrocyte sedimentation rate was 77 mm per hour (reference range, 0 to 20), and the blood level of C-reactive protein was 147.8 mg per liter (reference value, <8.0); other laboratory test results are shown in Table 1. Blood cultures were obtained. Tests for SARS-CoV-2 and human immunodeficiency virus types 1 and 2 were negative.

Dr. Eric M. Isselbacher: An ECG (Fig. 1A) showed sinus tachycardia, a borderline rightward axis, poor R-wave progression in leads V_1 through V_4, and nonspecific ST-segment and T-wave abnormalities. Transthoracic echocardiography (Fig. 1B through 1F and Videos 1, 2, and 3, available with the full text of this article at NEJM.org) revealed a mildly dilated left ventricle with diffuse hypokinesis and an ejection fraction of 34%. The aortic valve was tricuspid, without evidence of vegetation or aortic-root abscess, but there was incomplete closure of the aortic valve, which resulted in severe aortic regurgitation. The ascending thoracic aorta was dilated, measuring 44 mm in diameter, with thickened walls. The dilated left ventricle caused tethering of the mitral valve, which resulted in mild-to-moderate mitral regurgitation and mild left atrial enlargement. There was a moderate pericardial effusion without evidence of cardiac tamponade.

Dr. Vinit Baliyan: CT angiography of the chest, abdomen, and pelvis (Fig. 2) showed an irregular aortic contour with narrowing of the lumen in the vessels of the aortic arch and in the right renal artery. This imaging was performed with ECG gating, which allowed for the assessment of the aortic root. There was no evidence of a perivalvular abscess. However, circumferential wall thickening was present in the aorta at the root and in the ascending segment. In addition, there was circumferential wall thickening in partially visualized proximal vessels of the aortic arch and in the abdominal aorta. The mural thickening of the aorta showed enhancement that was suggestive of active inflammation; this enhancement was most visible in the abdominal aorta. Other findings included cardiomegaly with a markedly dilated left ventricle, alveolar pulmonary edema, a small pericardial effusion, and bilateral small pleural effusions.

Dr. Nestor: A diagnosis and management decisions were made.

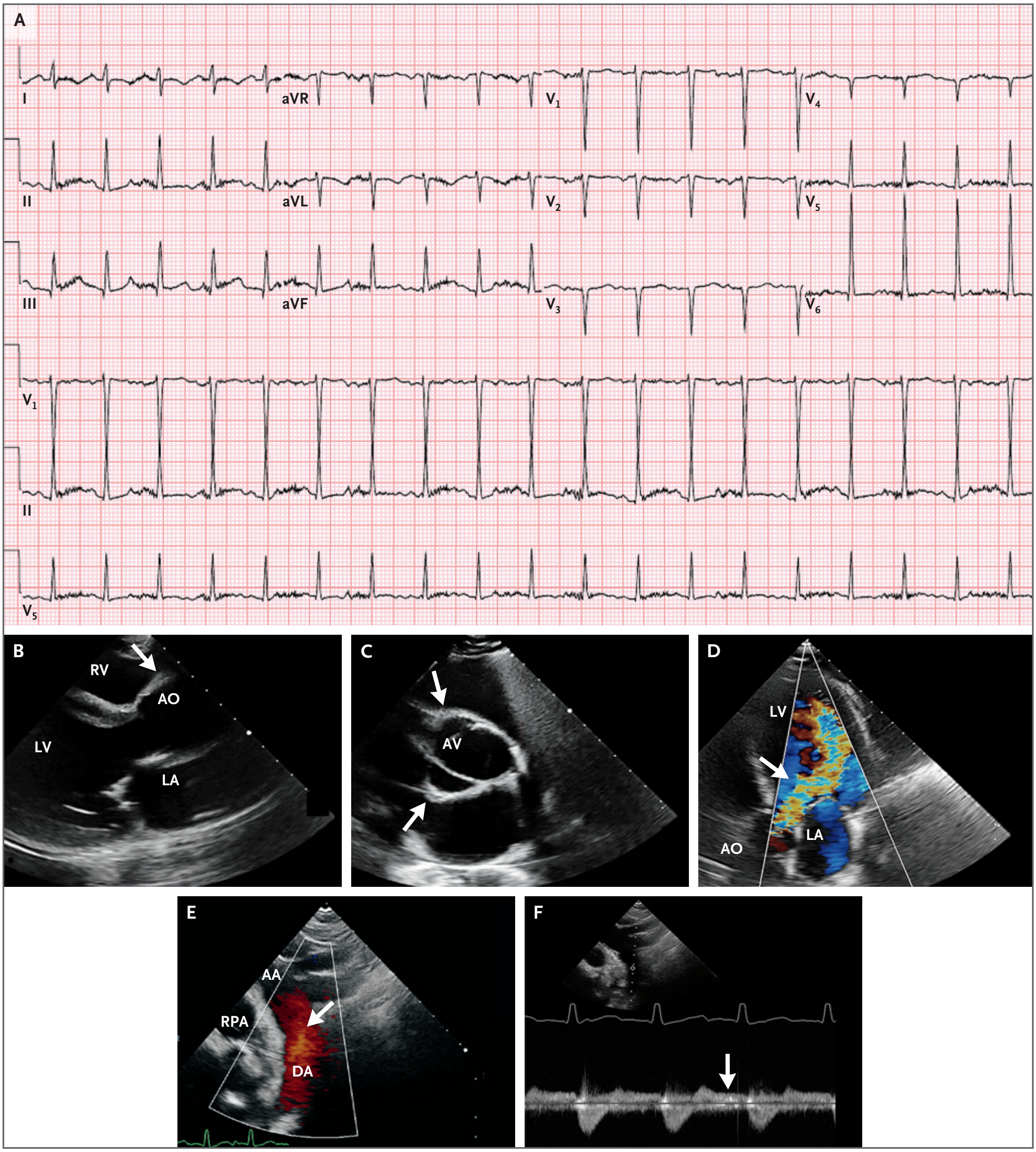
DIFFERENTIAL DIAGNOSIS

Dr. Cory A. Perugino: This young woman presented with a subacute illness involving the lungs, mitral and aortic valves, myocardium, pericardium, and aorta that was characterized by inflammatory features. The first step in developing a differential diagnosis is to consider whether a multiorgan inflammatory process caused this patient's presentation or whether a primary inflammatory process in the aorta could explain the valvular, cardiac, and pulmonary abnormalities.

Videos showing echocardiographic studies are available at NEJM.org

VALVULAR DISEASE

On echocardiography, both the aortic and mitral valves showed incomplete coaptation, but no vegetations or perforations were present and there was no leaflet thickening, which makes a thrombotic, infectious, or inflammatory valvular process unlikely. Instead, the patient's valvular abnormalities appear to be due to anatomical alterations in the adjacent tissues (i.e., left ventricular dilatation adjacent to the mitral valve and aortic-root dilatation adjacent to the aortic valve) that resulted in malcoaptation.



MYOCARDIAL DISEASE

The myocardium showed evidence of impaired function with reduced ejection fraction, but the troponin T level was not elevated. Therefore, ischemia and myocarditis are unlikely causes of myocardial disease in this young woman. Although other causes of cardiomyopathy should be considered, I suspect that the impaired left ventricular function was instead due to volume overload resulting from acute valvular damage.

PULMONARY DISEASE

Ground-glass opacities, which would have suggested a primary vasculitis of the lung, were not

Figure 1 (facing page). Cardiac Studies.
An electrocardiogram obtained at the time of the current presentation (Panel A) shows sinus tachycardia, a borderline rightward axis, poor R-wave progression in leads V_1 through V_4, and nonspecific ST-segment and T-wave abnormalities. On transthoracic echocardiography performed at the time of the current presentation, a parasternal long-axis view (Panel B) shows a dilated left ventricle (LV) and thickening of the aortic-root walls (arrow). A parasternal short-axis view of the aortic root (Panel C) shows a tricuspid aortic valve (AV) and thickening of the aortic-root walls (arrows). An apical five-chamber view (Panel D) shows a very large jet of severe aortic regurgitation (arrow). A suprasternal notch view with color Doppler (Panel E) shows holodiastolic flow reversal (arrow) in the descending thoracic aorta (DA). Continuous-wave Doppler tracing in the same suprasternal notch view (Panel F) confirms holodiastolic flow reversal (arrow), a finding that suggests severe aortic regurgitation. AA denotes aortic arch, AO aortic root, LA left atrium, RPA right pulmonary artery, and RV right ventricle.

observed on chest imaging, and the patient had no history of hemoptysis. Although she had had a recent SARS-CoV-2 infection, the combination of pulmonary edema and pleural effusions seen on chest imaging in the context of an elevated blood level of NT-proBNP and a reduced ejection fraction is most suggestive of pulmonary edema due to heart failure.

AORTIC DISEASE

Cross-sectional imaging of the aorta revealed circumferential wall thickening that was both continuous and extensive and involved the ascending thoracic aorta, aortic arch, descending thoracic aorta, and abdominal aorta, as well as major branch vessels. The circumferential nature of this finding is highly suggestive of aortitis; by contrast, the presence of irregular, asymmetric wall thickening would have suggested atherosclerotic aortic disease. The multifocal areas of stenosis without associated atherosclerotic plaque or calcification could indicate chronic inflammatory tissue damage leading to vessel-wall fibrosis.

The patient's presentation is most consistent with primary aortitis leading to valvular damage, myocardial dysfunction (heart failure), and pulmonary edema. The next step is to consider the causes of aortitis.

INFECTIOUS AORTITIS

In this patient with findings indicative of aortitis, it is important to consider whether the disorder is due to an infectious or noninfectious process. Although infectious aortitis is uncommon, it is imperative to rule out infectious causes, since the diagnosis of infection would drastically alter the management of the patient's illness and misdiagnosis would have devastating consequences.

The patient's symptoms began shortly after a dental procedure, which prompts consideration of an odontogenic portal of entry, followed by hematogenous spread and seeding of tissues. However, there was no evidence of valvular vegetations, perivalvular abscesses, or aortic-root abscesses that would suggest a primary infectious process. Moreover, the absence of fever in the context of such a fulminant presentation would be unusual if she had an infectious aortitis. Although the cross-sectional aortic imaging findings of circumferential wall thickening and contrast enhancement can occur with both infectious and noninfectious aortitis, it would be difficult to explain the continuous and extensive involvement of the aorta by infectious aortitis, which tends to be more regionally isolated.[1] Syphilitic aortitis, a feature of tertiary syphilis, is exceedingly rare and is generally confined to the thoracic aorta and not associated with multifocal stenoses or involvement of the major branch vessels of the aortic arch[2]; therefore, this diagnosis is highly unlikely in this patient.

NONINFECTIOUS AORTITIS

Noninfectious aortitis is most often caused by a primary large-vessel vasculitis such as giant-cell arteritis or Takayasu's arteritis. Less often, noninfectious aortitis is caused by a systemic autoimmune disease process such as IgG4-related disease, rheumatoid arthritis, spondyloarthritis, and others.[3]

Giant-Cell Arteritis

Giant-cell arteritis is the most common primary large-vessel vasculitis affecting the aorta and is characterized by granulomatous inflammation that typically affects the muscular layer, although all layers of the aorta can be inflamed.[4] The presenting features of giant-cell arteritis vary depending on the vessels involved and can include localized headache, scalp tenderness, vision loss, and jaw claudication, as well as myalgias and symmetric stiffness of the shoulder and pelvic girdle (i.e., polymyalgia rheumatica).[5]

On the basis of classification criteria,[6] the minimum age at which giant-cell arteritis is diagnosed is 50 years, and the median age is in the seventh or eighth decade of life.[7-10] Therefore, giant-cell arteritis is not a consideration in this 19-year-old patient.

Takayasu's Arteritis

Takayasu's arteritis is the most common large-vessel vasculitis affecting the aorta among patients younger than 50 years of age, and it typically occurs in women younger than 40 years of age. Takayasu's arteritis is characterized by a

Figure 2 (facing page). CT Angiography of the Chest, Abdomen, and Pelvis.

CT angiography was performed at the time of the current presentation. A curved planar reformatted image obtained during the arterial phase (Panel A) shows an irregular contour of the descending aorta (yellow bracket). Maximum-intensity-projection images (Panels B and C) show luminal narrowing of partially visualized vessels of the aortic arch, including a mild focal stenosis of the innominate artery and diffuse narrowing of the proximal bilateral common carotid arteries (severe on the right side and mild to moderate on the left side) (Panel B, arrows); a severe focal stenosis is seen in the right renal artery (Panel C, arrow). Axial images obtained during the arterial phase (Panels D, E, and F) show circumferential wall thickening in the aorta at the root (Panel D, arrow) and in the ascending segment (Panel E, arrow). Circumferential wall thickening is also present in partially visualized proximal vessels of the aortic arch (Panel F, arrows). An axial image obtained during the venous phase (Panel G) shows circumferential wall thickening in the abdominal aorta (arrow) and corresponding mural enhancement.

transmural granulomatous infiltrate.[4] The ascending aorta and vessels of the aortic arch are classically affected, but five categories of aortic involvement have been established on the basis of angiographic distribution.[11] Long segments of contiguous involvement as well as multiple areas of stenosis or frank vessel occlusion, as seen in this patient, are typical findings associated with Takayasu's arteritis; vessel damage with aneurysm formation can also occur. Patients often present with symptoms related to distal hypoperfusion of the affected vessels, including arm claudication, syncope, stroke, or diminished or absent pulses on physical examination.[12,13] However, aortic-root aneurysm with secondary aortic regurgitation is also a classic complication of Takayasu's arteritis.[14] Takayasu's arteritis is a possible explanation for this patient's presentation, but there are several other diagnoses to consider.

Systemic Autoimmune Disease

IgG4-related disease is a multiorgan disease that is characterized by the insidious accumulation of immune cells in organs, which generally manifests as diffuse organ enlargement. Typical organ involvement includes the lacrimal glands, salivary glands, pancreas, bile ducts, retroperitoneal tissues, kidneys, and aorta.[15] IgG4-related disease most often develops in the sixth decade of life and has male predominance, which makes this diagnosis unlikely in this young female patient.[16,17] However, IgG4-related disease has been reported across various age groups, including pediatric patients.[18] Large-vessel vasculitis is a well-established manifestation of IgG4-related disease, and affected patients may have either thoracic or abdominal aortitis with or without aneurysm formation.[19] The patient had a combination of thoracic and abdominal aortic involvement as well as multifocal large-vessel stenoses, findings that would be uncommon in a patient with IgG4-related disease. Moreover, the patient's fulminant presentation, subacute evolution of disease, and demographic characteristics make IgG4-related disease an unlikely diagnosis.

Other systemic autoimmune diseases can cause aortitis, including systemic lupus erythematosus, rheumatoid vasculitis, sarcoidosis, Cogan's syndrome, Behçet's disease, ankylosing spondylitis, granulomatosis with polyangiitis, and relapsing polychondritis. However, these diseases are considered to be secondary causes of large-vessel vasculitis and generally require the presence of associated disease features that were not observed in this patient.

SUMMARY

Takayasu's arteritis is the most likely explanation for this patient's subacute, multiorgan disease presentation with inflammatory features. I suspect that she had aortitis complicated by aortic-root dilatation that led to aortic regurgitation and heart failure.

CLINICAL IMPRESSION

Dr. Isselbacher: On the basis of the subacute presentation of this patient's illness, the circumferential thickening of the walls of the aorta and its branches, and the lack of evidence of endocarditis, our clinical impression was aortitis, with Takayasu's arteritis being the most likely diagnosis. We hypothesized that the dilatation of the ascending thoracic aorta and the thickening of the aortic-valve cusps led to incomplete aortic-valve closure resulting in severe aortic regurgitation, which in turn led to subacute left ventricular dilatation and systolic dysfunction and culminated in pulmonary edema, tachycardia, and tachypnea.

CLINICAL DIAGNOSIS

Takayasu's arteritis.

DR. CORY A. PERUGINO'S DIAGNOSIS

Takayasu's arteritis.

INITIAL CARDIOVASCULAR MANAGEMENT

Dr. Isselbacher: The patient was treated with supplemental oxygen and intravenous diuretic agents, but tachypnea persisted and hypotension developed. In the context of clinical instability due to severe aortic regurgitation, cardiac surgery for aortic-valve replacement was performed on hospital day 3. The native aortic-valve cusps were thickened and retracted, and the aorta was only mildly dilated. A 23-mm pericardial valve was used for aortic-valve replacement; this valve was chosen over a mechanical prosthetic valve because the latter would require the use of warfarin anticoagulation, which would complicate the patient's plans to have children within the next 10 years. Transcatheter aortic-valve replacement was considered to be a poor option given that the available devices are not designed or approved for use in the treatment of pure aortic regurgitation and can potentially dislodge.

PATHOLOGICAL DISCUSSION

Dr. Rex N. Smith: A surgical specimen of the aortic valve and a biopsy specimen of the aorta (Fig. 3) were submitted for pathological evaluation. Neither specimen had pathological features that were characteristic of an inflammatory or infectious process. The aortic-valve specimen showed myxomatous degeneration. The biopsy specimen obtained from the aorta was considered to be normal, but aortitis could not be ruled out because of the small amount of tissue that had been obtained.

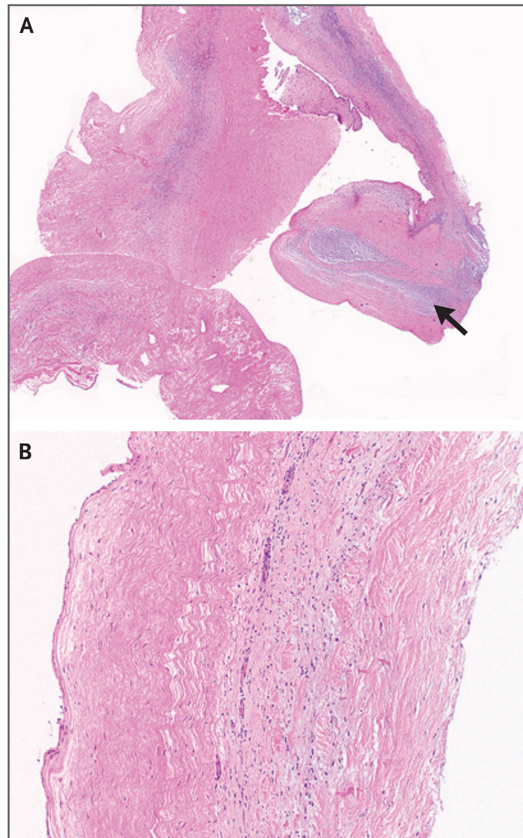


Figure 3. Surgical and Biopsy Specimens.
Hematoxylin and eosin staining of a fragment of the aortic valve (Panel A) shows myxomatous degeneration (arrow). Hematoxylin and eosin staining of a biopsy specimen of the aorta (Panel B) does not show evidence of inflammation.

RHEUMATOLOGIC DISCUSSION

Dr. Sebastian H. Unizony: Takayasu's arteritis is a rare disorder, with a prevalence of 0.9 to 2.6 cases per million persons in the United States and 40 cases per million persons in Japan.[20,21] The diagnosis of Takayasu's arteritis is often delayed owing to the rarity of the disorder and the heterogeneity of its clinical presentation.

The onset of Takayasu's arteritis is generally subacute or insidious, and disease severity ranges from asymptomatic to catastrophic, the latter of which was seen in this patient.[22] Approximately one third of patients have constitutional or musculoskeletal signs and symptoms such as fatigue, fever, weight loss, myalgias, and arthralgias,[23] and two thirds of patients present with manifestations associated with arterial involvement, most commonly in the aorta and the subclavian, carotid, and renal arteries and less often in the vertebral, pulmonary, coronary, mesenteric, and iliofemoral arteries.[23,24]

Patients with Takayasu's arteritis often have arm claudication.[22,25] Auscultation of the anterior aspect of the neck and the supraclavicular region can reveal characteristic bruits reflecting stenosis in the carotid and subclavian arteries, as was seen in this patient.[22] Other physical findings may include abdominal bruits (due to renal artery stenosis), pulse deficits (e.g., deficits of the radial pulse), and blood pressure measurements that differ by more than 10 to 20 mm Hg between arms.

Cardiac disease is a major cause of complications and death in patients with Takayasu's arteritis.[24,26] Some degree of aortic regurgitation is detected in 15 to 50% of affected patients and primarily occurs as a consequence of the separation of the aortic-valve leaflets due to aortic-root dilatation.[26] The clinical presentation of this complication ranges from clinically silent to decompensated heart failure, the latter of which was seen in this patient. Other potential causes of left ventricular dysfunction in patients with Takayasu's arteritis include pressure overload due to hypertension associated with renal artery stenosis, myocardial ischemia due to coronary arteritis, and myocarditis.[26] Some patients have pulmonary hypertension and pain overlying the blood vessels, including in the neck and back, as reported by this patient; the vascular-related pain (angiodynia) is probably due to the stimulation of nociceptive receptors along the inflamed arteries (e.g., the carotid arteries or the aorta).[24]

Most patients with Takayasu's arteritis have nonspecific laboratory abnormalities that reflect systemic inflammation, including anemia, elevation of the erythrocyte sedimentation rate and C-reactive protein level, and thrombocytosis,[24,27] all of which were seen in this patient. Confirmation of the Takayasu's arteritis diagnosis in a patient with a clinical syndrome suggestive of this disorder requires vascular imaging or biopsy. Because tissue from large arteries is obtained from only a minority of patients undergoing surgery, vascular imaging is the test most frequently used to establish the diagnosis.[28] When biopsy is feasible, findings reveal different degrees of arterial inflammation (with infiltrate composed of lymphocytes, macrophages, and giant cells) and remodeling such as fibrosis, neoangiogenesis, and intimal hyperplasia.[4] The sensitivity and specificity of arterial biopsy for the diagnosis of Takayasu's arteritis have not been defined. In this patient, the negative biopsy results could have been caused by the sampling of an uninvolved aortic segment. This patient's clinical syndrome and findings on vascular imaging studies were most consistent with a diagnosis of Takayasu's arteritis.

DISCUSSION OF MANAGEMENT

Dr. Unizony: The management of Takayasu's arteritis is challenging owing to the absence of highly accurate biomarkers to monitor disease activity, the possibility of subclinical vascular inflammation, and most importantly, the paucity of randomized, controlled trials to guide therapy.

Glucocorticoid therapy is effective for inducing remission when used at moderate-to-high doses. After 2 to 4 weeks of treatment at the initial dose, glucocorticoids are generally tapered over a period of 6 to 12 months.[29] However, longer courses of low-dose glucocorticoids are sometimes needed to maintain remission. Unfortunately, when glucocorticoids are prescribed as monotherapy, relapses and glucocorticoid-induced toxic effects occur in most patients.[12,13,25,30-32] In patients presenting with organ-threatening or life-threatening manifestations, like this patient, intravenous pulse glucocorticoid therapy may be warranted. In this patient, treatment with high-dose intravenous glucocorticoids was initiated on hospital day 2.

Surgeons and interventional radiologists play important roles in the treatment of Takayasu's arteritis. Surgical procedures are performed in approximately 50% of patients.[22,23,32,33] Surgery can be lifesaving for patients with severe aortic regurgitation, large aortic aneurysms, or aortic dissection. Revascularization (e.g., percutaneous angioplasty with or without stenting or bypass surgery) is performed in patients with refractory hypertension caused by renal artery stenosis and in patients with coronary, cerebral, mesenteric, or limb ischemia that is symptomatically moderate or severe despite medical therapy.[33] Whenever possible, surgical procedures are deferred until immunomodulatory therapy has suppressed the arterial inflammation and daily prednisone doses have been reduced below 7.5 mg owing to concerns about surgical complications associated with operating on friable inflamed tissues and poor wound healing, respectively. This patient received high-dose intravenous glucocorticoids for 2 days before undergoing aortic-valve replacement, but further delay was not possible in

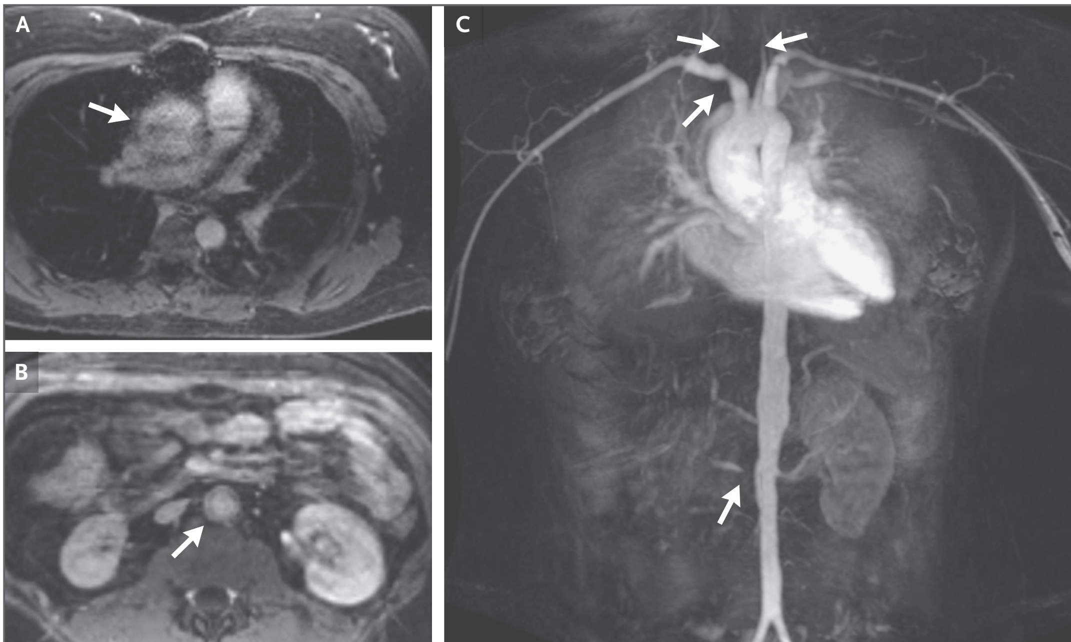


Figure 4. **Magnetic Resonance Angiography 4 Months after Surgery.**

Axial contrast-enhanced T1-weighted fat-saturated images show aortic-wall thickening in the ascending segment (Panel A, arrow) and in the abdominal segment (Panel B, arrow). The image quality is somewhat limited with respect to the ascending aorta owing to susceptibility artifacts from sternal wires. A coronal maximum-intensity-projection image (Panel C) shows persistent luminal narrowing in the vessels of the aortic arch and in the right renal artery (arrows); there is poor visualization of the right common carotid artery on this image owing to severe diffuse narrowing.

the context of clinical instability due to acute severe aortic regurgitation, which necessitated immediate surgery.

In view of the high incidence of relapse associated with glucocorticoid monotherapy in the treatment of Takayasu's arteritis, nontargeted immunosuppressants (e.g., methotrexate, azathioprine, and mycophenolate mofetil) have been used in an attempt to maintain remission and decrease glucocorticoid exposure.[13,25,27,30-36] Although treatment with a combination of glucocorticoids and nontargeted immunosuppressants has been reported to result in better outcomes than treatment with glucocorticoids alone, relapse and silent disease progression continue to occur in some patients.[37,38] Moreover, data from randomized clinical trials regarding the safety and efficacy of nontargeted immunosuppressants in the treatment of Takayasu's arteritis are lacking.

Results from uncontrolled studies and a phase 2 randomized clinical trial suggest that tumor necrosis factor α (TNF-α) inhibitors (e.g., infliximab and adalimumab)[37-42] and interleukin-6 inhibitors (e.g., tocilizumab)[41-43] are more efficacious than nontargeted immunosuppressants in maintaining remission, sparing glucocorticoid use, and reducing accrual of new arterial lesions in patients with Takayasu's arteritis. Until further evidence of the benefits associated with various pharmacologic interventions is available from randomized clinical trials, the 2021 American College of Rheumatology–Vasculitis Foundation guideline for the management of Takayasu's arteritis conditionally recommends the use of nontargeted immunosuppressants or TNF-α inhibitors in combination with glucocorticoids in patients with active Takayasu's arteritis (including new-onset disease) and tocilizumab in patients who do not have an adequate response to the aforementioned therapies.[29]

Dr. Nestor: In this patient, treatment with oral glucocorticoids was started postoperatively, with a plan to taper the dose over a period of 6 to 9 months. While we awaited insurance authorization for treatment with a TNF-α inhibitor, glucocorticoid-sparing treatment with weekly methotrexate and folic acid was started. The patient's treatment was eventually switched to adalimumab. A repeat echocardiogram obtained 1 month after aortic-valve replacement was notable for normalization of left ventricular function. Dyspnea resolved, and the patient was able to return to work 2 months after surgery. Four months after surgery, she began regular monitoring with magnetic resonance angiography (Fig. 4).

Dr. Isselbacher: Because the patient received a biologic aortic-valve prosthesis at a young age, the risk of accelerated degeneration of the prosthesis is high; therefore, the valve will need to be monitored on a regular basis. Physical examinations and echocardiography are planned to evaluate for the presence of valve dysfunction. The patient's aorta and its branches are at risk for further remodeling with either stenotic or aneurysmal lesions; surveillance imaging is important to identify any lesions that may merit intervention.

FINAL DIAGNOSIS

Takayasu's arteritis.

This case was presented at Rheumatology Grand Rounds.
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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